INCH-POUND

MIL-P-83461/1B
8 May 1989
SUPERSEDING
MIL-P-83461/1A
8 September 1978

MILITARY SPECIFICATION SHEET

PACKING, PREFORMED, PETROLEUM HYDRAULIC FLUID RESISTANT IMPROVED PERFORMANCE AT 275°F (135°C) SIZES AND TOLERANCES

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation MIL-P-83461

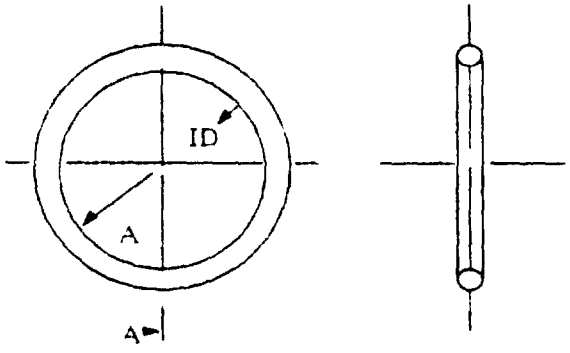
PURPOSE The purpose of this specification sheet is to set up a standardized part numbering system for 0-rings procured procured to MIL-P-83461

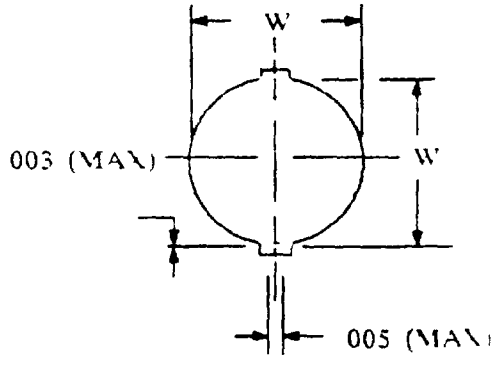
(B) APPLICABLE DOCUMENT

AS 568 Aerospace Size Standarde for O-Rings

(Applications for copies should be addressed to Society of Automotive Engineers 400 Commonwealth Drive Warrendale, PA 15096)

(B) DIMENSIONS AND TOLERANCES





ALL DIAMETERS OF CROSS-SECTION SHALL EQUAL W
SECTION A - A

(B) denotes changes

AMSC N/A FSC 5330

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(B) PART NUMBERS When this document is used for procurement of O-rings part numbers shall consist of the following

a The letter M and the specification sheet number

b A dash followed by the appropriate dash number from table I or table II

Example 1 (table I)	Example 2 (table II)
M83461/1-214	M83461/1-337

(B) INTENDED USE O-rings listed in table I are recommended for use in packing glands as specified in MIL-G-5514 O-rings listed in table II are the remainder of those listed in AS568, however, the aplicable gland sizes for these O-rings are not specified in MIL-G-5514 The table II sizes are not recommended for hydraulic applications

Custodians
Army - AV
Navy - AS
Air Force - 11

Review activities
Air Force - 82, 99
DLA - IS

User activities
Army - ME
Air Force - 71

Preparing activity
Air Force - 11

Project No 5330-0797

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TABLE I Sizes listed in MIL-G-5514

DASH NO 1/	ACTUAL SIZE				DASH NO 1/	ACTUAL SIZE			
	WIDTH		INSIDE DIAMETER			WIDTH		INSIDE DIAMETER	
001	0 040	±0 003	0 029	±0 004	118	0 103	±0 003	0 862	±0 010
002	0 050		0 042		119			0 924	
003	0 060		0 056		120			0 987	
004	0 070	±0 003	0 070	±0 005	121			1 049	
005			0 101		122			1 112	
006			0 114		123			1 174	±0 012
007			0 145		124			1 237	
008			0 176		125			1 299	
009			0 208		126			1.362	
010			0 239		127			1 424	
011			0 301		128			1 487	
012			0 364		129			1 549	±0 015
013			0 426		130			1 612	
014			0 489		131			1 674	
015			0 551	±0 007	132			1 737	
016			0 614	±0 009	133			1 799	
017			0 676		134			1 862	
018			0 739		135			1 925	±0 017
019			0 801		136			1 987	
020			0 864		137			2 050	
021			0 926		138			2 112	
022			0 989	±0 010	139			2 175	
023			1 051		140			2 237	
024			1 114		141			2.300	±0.020
025			1 176	±0 011	142			2 362	
026			1 239		143			2 425	
027			1 301		144			2 487	
028			1 364	±0 013	145			2 550	
					146			2 612	
110	0 103	±0 003	0 362	±0 005	147			2 675	±0 022
111			0 424		148			2 737	
112			0 487		149			2.800	
113			0 549	±0 007					
114			0 612	±0 009	210	0 139	±0 004	0 734	±0 010
115			0 674		211			0 796	
116			0 737		212			0 859	
117			0 799	±0 010	213			0 921	

1/ All dash numbers conform with the uniform dash system of AS568

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TABLE I Sizes listed in MIL-G-5514 - Continued

DASH NO 1/	ACTUAL SIZE				DASH NO 1/	ACTUAL SIZE			
	WIDTH		INSIDE DIAMETER			WIDTH		INSIDE DIAMETER	
214	0 139	±0 004	0 984	±0 010	325	0 210	±0 005	1.475	±0 015
215			1 046		326			1.600	
216			1 109	±0 012	327			1 725	
217			1.171		328			1 850	
218			1 234		329			1 975	±0 018
219			1 296		330			2 100	
220			1 359		331			2 225	
221			1 421		332			2 350	
222			1 484	±0 015	333			2 475	±0 020
223			1 609		334			2 600	
224			1 734		335			2 725	
225			1 859	±0 018	336			2 850	
226			1 984		337			2 975	±0 024
227			2 109		338			3 100	
228			2 234	±0 020	339			3 225	
229			2 359		340			3 350	
230			2 484		341			3 475	
231			2 609		342			3 600	±0 028
232			2 734	±0 024	343			3 725	
233			2 859		344			3.850	
234			2 984		345			3 975	
235			3 109		346			4 100	
236			3 234		347			4 225	±0 030
237			3 359		348			4 350	
238			3 484		349			4 475	
239			3 609	±0 028					
240			3 734		425	0 275	±0 006	4 475	±0 033
241			3 859		426			4 600	
242			3 984		427			4 725	
243			4 109		428			4 850	
244			4 234	±0 030	429			4 975	±0 037
245			4 359		430			5.100	
246			4 484		431			5.225	
247			4 609		432			5 350	

NOTE O-ring sizes -013 through -028, -117 through -149 and -223 through -247 are intended only for use as static seals, and are not to be used in applications involving reciprocating or rotary movement

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(B) TABLE II Sizes not listed in MIL-G-5514

DASH NO 1	ACTUAL SIZE WIDTH		ACTUAL SIZE INSIDE DIAMETER		DASH NO 1	ACTUAL SIZE WIDTH		ACTUAL SIZE INSIDE DIAMETER	
029	0 070	±0 003	1 489	±0 013	155	0 103	±0 003	3 987	±0 028
030			1 614		156			4 237	±0 030
031			1 739	±0 015	157			4.487	
032			1 864		158			4.737	
033			1 989	±0 018	159			4.987	±0 035
034			2 114		160			5 237	
035			2 239		161			5 487	
036			2 364		162			5 737	
037			2 489		163			5 987	
038			2 614	±0 020	164			6 237	±0 040
039			2 739		165			6 487	
040			2 864		166			6 737	
041			2 989	±0 024	167			6 987	
042			3 239		168			7 237	±0 045
043			3 489		169			7 487	
044			3 739	±0 027	170			7 737	
045			3 989		171			7 987	
046			4 239	±0 030	172			8 237	±0 050
047			4 489		173			8 487	
048			4 739		174			8 737	
049			4 989	±0 037	175			8 987	
050			5 239		176			9 237	±0 055
					177			9 487	
102	0 103		0 049	±0 005	178			9 737	
103			0 081						
104			0 112		201	0 139	±0 004	0 171	±0 005
105			0 143		202			0 234	
106			0 174		203			0 296	
107			0 206		204			0 359	
108			0 237		205			0 421	
109			0 299		206			0 484	
					207			0 546	±0 007
150			2 862	±0 022	208			0 609	±0 009
151			2 987	±0 024	209			0 671	
152			3 237						
153			3 487		248			4 734	±0 030
154			3 737	±0 028	249			4 859	±0 035

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TABLE I Sizes listed in MIL-G-5514 - Continued

DASH NO 1	ACTUAL SIZE			
	WIDTH		INSIDE DIAMETER	
433	0 275	±0 006	5 475	±0.037
434			5 600	
435			5 725	
436			5 850	
437			5 975	
438			6 225	±0.040
439			6 475	
440			6 725	
441			6 975	
442			7 225	±0.045
443			7 475	
444			7 725	
445			7 975	
446			8 475	±0 055
447			8 975	
448			9 475	
449			9 975	
450			10 475	±0 060
451			10 975	
452			11.475	
453			11 975	
454			12 475	
455			12 975	
456			13 475	±0 070
457			13.975	
458			14 475	
459			14 975	
460			15 475	

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(B) TABLE II Sizes not listed in MIL-G-5514 - Continued

DASH NO 1	ACTUAL SIZE WIDTH		ACTUAL SIZE INSIDE DIAMETER		DASH NO 1	ACTUAL SIZE WIDTH		ACTUAL SIZE INSIDE DIAMETER	
250	0 139	±0 004	4 984	±0.035	309	0.210	±0 005	0 412	±0 005
251			5 109		310			0 475	
252			5 234		311			0 537	±0.007
253			5 359		312			0 600	±0 009
254			5 484		313			0 662	
255			5 609		314			0 725	±0 010
256			5 734		315			0 787	
257			5 859		316			0 850	
258			5 984		317			0 912	
259			6 234	±0 040	318			0 975	
260			6 484		319			1 037	
261			6 734		320			1 100	±0.012
262			6 984		321			1 162	
263			7 234	±0 045	322			1 225	
264			7 484		323			1 287	
265			7 734		324			1 350	
266			7 984						
267			8 234	±0 050	350			4 600	±0 030
268			8 484		351			4 725	
269			8 734		352			4 850	
270			8 984		353			4.975	±0 037
271			9 234	±0 055	354			5.100	
272			9 484		355			5 225	
273			9 734		356			5 350	
274			9 984		357			5 475	
275			10 484		358			5 600	
276			10 984	±0 065	359			5 725	
277			11 484		360			5 850	
278			11 984		361			5 975	
279			12.984		362			6 225	±0 040
280			13 984		363			6 475	
281			14 984		364			6.725	
282			15 955	±0 075	365			6.975	
283			16 955	±0 080	366			7.225	±0 045
284			17 955	±0 085	367			7.475	

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(B) TABLE II Sizes not listed in MIL-G-5514 - Continued

DASH NO 1	ACTUAL SIZE WIDTH		INSIDE DIAMETER	
368	0 210	±0 005	7 725	±0 045
369			7 975	
370			8.225	±0 050
371			8 475	
372			8 725	
373			8 975	
374			9 225	±0 055
375			9 475	
376			9 725	
377			9 975	
378			10 475	±0 060
379			10 975	
380			11 475	±0.065
381			11 975	
382			12 975	
383			13 975	±0 070
384			14 975	
385			15.955	±0 075
386			16 955	±0 080
387			17 955	±0 085
388			18 955	±0 090
389			19 955	±0 095
390			20 955	
391			21 955	±0 100
392			22 940	±0 105
393			23 940	±0 110
394			24 940	±0 115
395			25 940	±0 120
461	0 275	±0 006	15 955	±0 075
462			16 455	
463			16 955	±0 080
464			17 455	±0 085
465			17 955	
466			18 455	
467			18 955	±0 090
468			19 455	

DASH NO 1	ACTUAL SIZE WIDTH		INSIDE DIAMETER	
469	0 275	±0 006	19 955	±0 095
470			20 955	
471			21.955	±0 100
472			22.940	±0 105
473			23 940	±0 110
474			24 940	±0.115
475			25.940	±0 120

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